Capstone 2: Final Report

Evan Luu

Aug 24, 2026

Overview and Results

Everyday, every hour, every minute statistically speaking, a motor accident will occur. Much to the chagrin of the people involved and affected, it seems like there is no way of permanently solving this problem outside of removing drivers from the road. However what if we were able to predict the likely causes, it would be a boon for everyone on the road and the insurance actuaries. Sadly though, an answer to that is rarely simple as it is a mixture of both drivers and their equipment. Hence, the need for a data science / machine learning model to help determine a method to predict who is the most likely perpetrators.

However in the end though, this deployable model will not be possible as of right now. As right now, the issues that come with this model is that it is simply not accurate enough to be trusted. Given hundreds of thousands of rows, and about a dozen features to help determine an answer of who is at fault for an accident, this proved to only come up with an accuracy score of about 60%. Which is far from the industry standard around 80% for to be considered deployable. Shown below are my findings from the model testing, as in figure 1 and figure 2. The main takeaway from these matrices is that currently, there is an inherent bias to assume that someone did not cause an accident, in this case 0. Which is highly concerning since it appears to misclassify a huge portion of the testing sample as not being responsible. Hence for future revisions, the training should be revisited to see if there are ways to fix this inherent bias in both decision tree / random forest models. This is also inherently seen in other model types as well such as linear regression seen in figure 3.

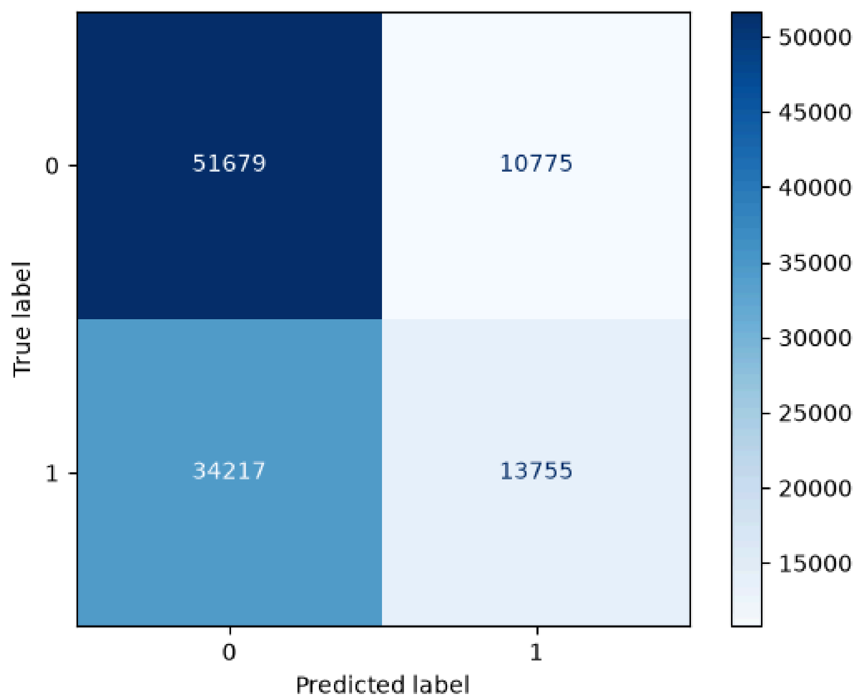

Figure 1: Gini Predict Model Confusion Matrix

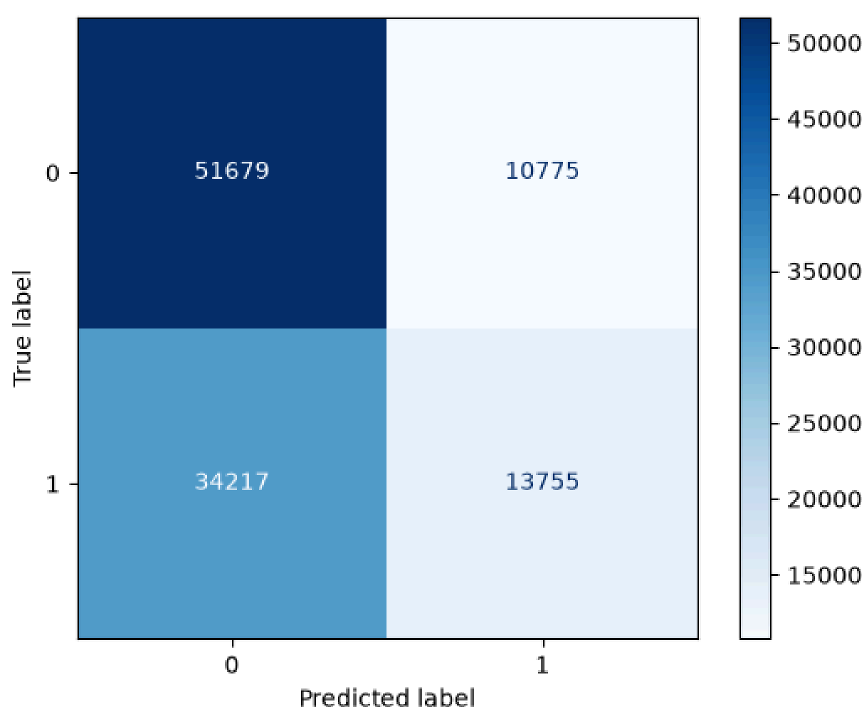

Figure 2: Entropy Model Confusion Matrix

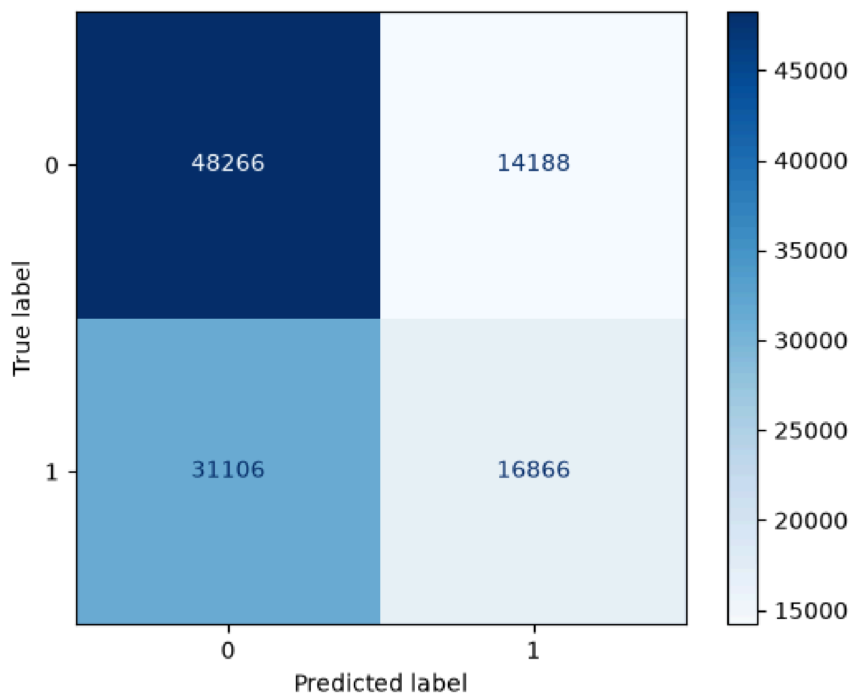

Figure 3: Logistic Regression Confusion Matrix

Recommendations

For current recommendations, I cannot recommend using these models in these current states. As more time is needed to understand and refine these models for production usage. However, what I can recommend is that for future revisions, here are the following factors to focus on: popular car models, speed limits and traffic violations. The reason being that the majority of these values make up the vast majority of traffic incidents in this dataset. Along with it helps paint a picture that shows that car size does seem to make some impact on the accident at-fault responsibility rate. Some figures are shown below to help illustrate these recommendations (Fig 4-8).

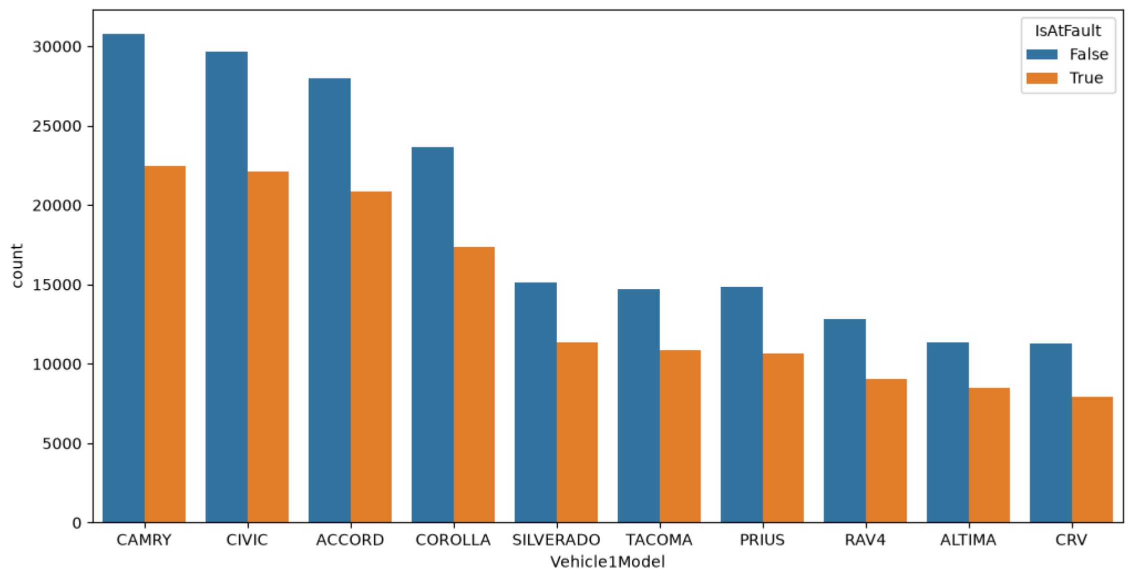

Fig 4: Top 10 Vehicles in accidents

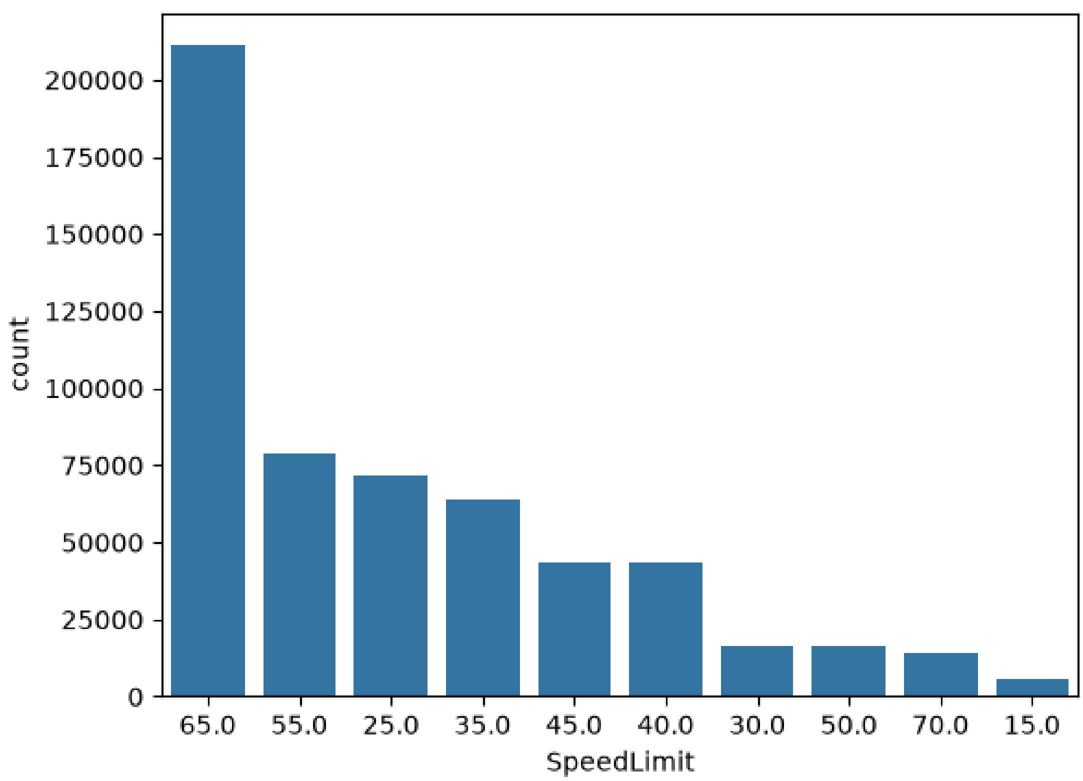

Fig 5: Speed limits at accident sites

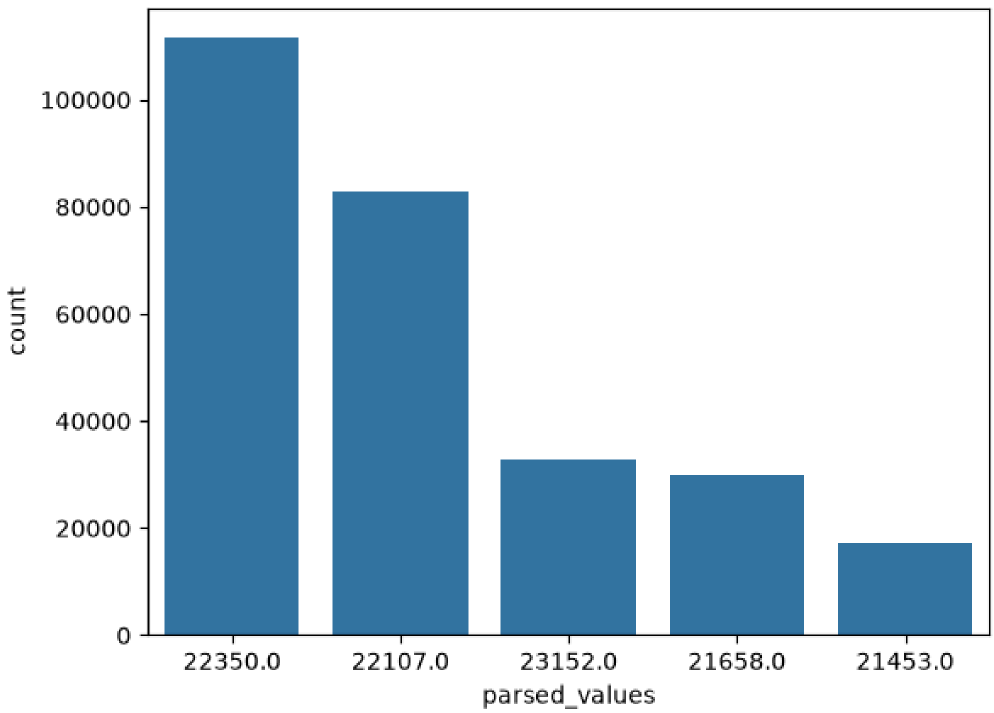

Fig 6: Traffic violation codes leading to accident

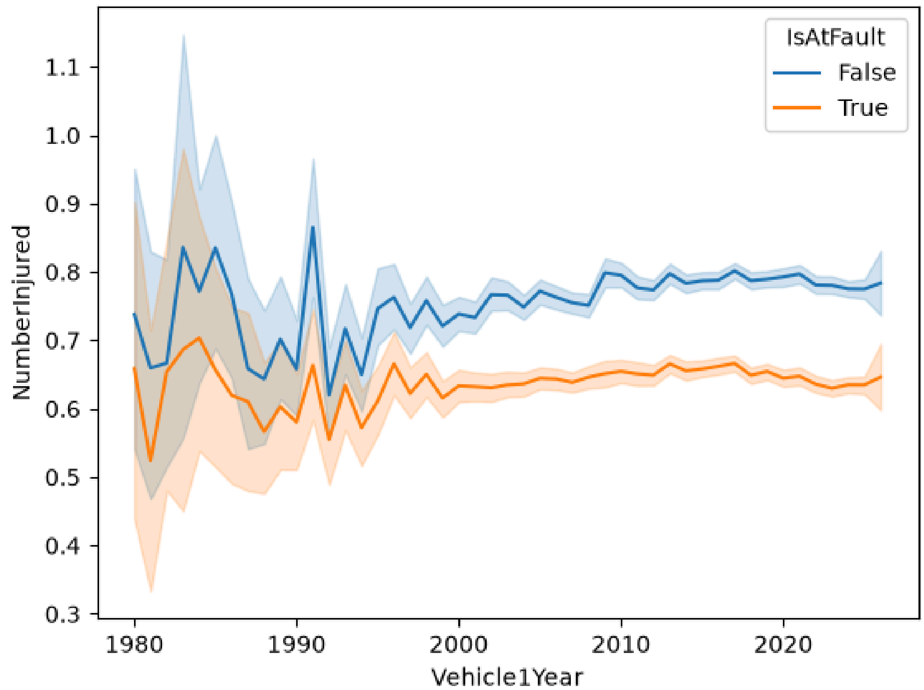

Fig 7: Trends of Injuries in Accidents Based on Accountability

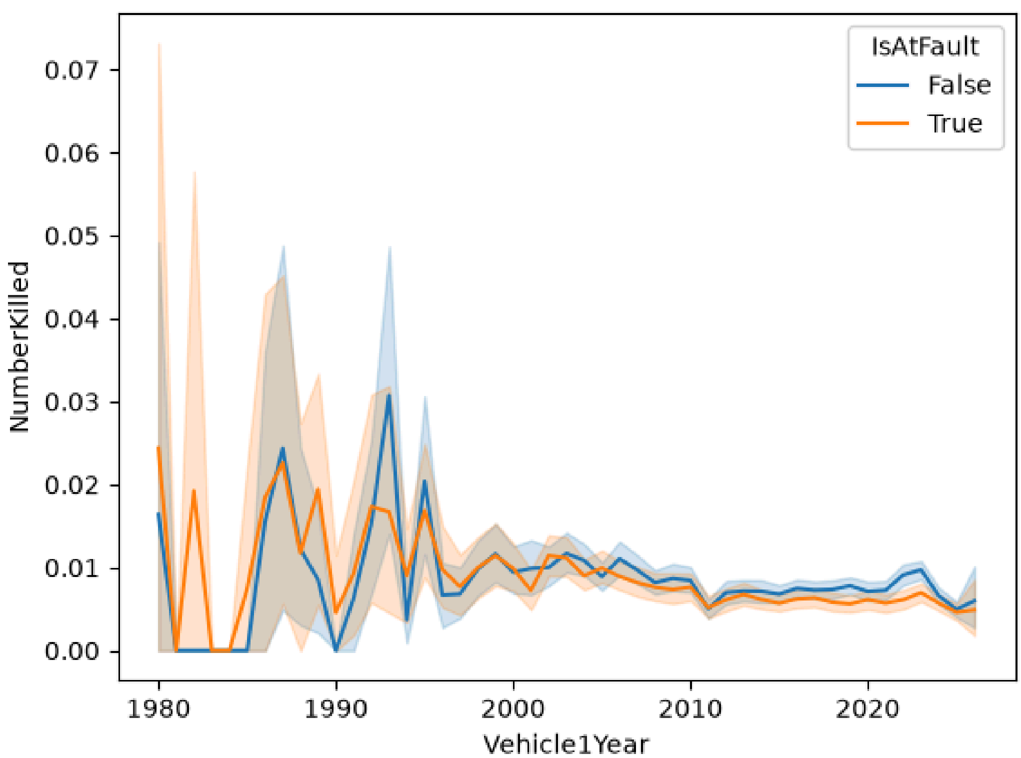

Fig 8: Trends of Fatalities in Accidents Based on Accountability